
PROJECT PROPOSAL

AI-Powered Visual Search for Zen Cart Locomotive Inventory System

Prepared for	Train Collection Website, USA	Prepared by	Ahmed Ali, Datix AI
Date	August 7, 2026	Timeline	10–14 days

1. Project Overview

Your Zen Cart site (PHP/MySQL) runs as a private inventory and preparation-request system for your train collection browsing only, no checkout. Visitors add locomotives/rolling stock to a cart and submit the list by email for you to prepare. Locomotive lookup currently works via QR code stickers linking to each product page.

This project adds Visual Search: pointing a phone camera at a locomotive to automatically identify it and open its product page, as an additional lookup method alongside the existing QR system (which stays in place as the reliable fallback).

2. Scope of Work

Included

- Review of the existing Zen Cart codebase, database schema, and product photo set.
- Setup of a cloud image-recognition project and training of a custom classifier on your locomotive photos, tagged by SKU/product ID.
- A new camera-upload page built directly into the existing Zen Cart site — no rebuild, no platform migration.
- Integration logic: photo uploaded → sent to the AI service → best-match product ID(s) returned → redirect to the matching product page.
- A "Top matches" fallback screen for low-confidence results — expected for near-identical locomotives.
- Real-world testing with actual phone photos (not just clean training images) and threshold tuning.
- Short handover documentation on adding new locomotives to the visual search index going forward.

Not included

- Migration to WordPress/WooCommerce or any other platform.
- Re-enabling checkout or credit card processing.
- Guaranteed accuracy on near-identical locomotives (see Limitations).
- Ongoing cloud API usage costs beyond setup billed separately, see Section 5.

3. Timeline

1. Access & discovery (Day 1–2): review site/database access and existing photo set.
2. Photo dataset prep (parallel, client-side): reshoot locomotives on solid background, consistent lighting, 2–3 angles each.
3. Model setup & training (Day 2–5): tag and upload images, train, review per-item accuracy.
4. Model refinement (Day 5–7): add extra images for confused/similar locomotive pairs, retrain.
5. Zen Cart integration (Day 6–9): build the upload page, wire to the AI service and product database.
6. Testing & tuning (Day 9–12): real phone-photo testing, confidence threshold tuning.
7. Review & handover (Day 12–14): walkthrough, adjustments, documentation delivery.

Total: 10–14 days from receiving reshoot photos and site access. Calendar time may extend depending on photo turnaround.

4. Development Fee

Item	Amount (USD)
Visual search development — full scope above	\$300-350
Final figure confirmed after reviewing photo set & inventory size	Within range above

Flat project fee, not hourly. Testing fee will be charged hourly. It can take 4 to 5 hours.

5. Cross-Border Payment & Working Arrangement

- Any cloud AI service account (see Section 6) should be created and billed under your own name/card, so usage fees are visible to you directly with no markup.
- Remote access, if needed for site/server work: the same tools already used on other projects (AnyDesk, or your host's cPanel/FTP access) no additional software required on your end.

6. Hosting the AI Model Free vs. Paid Options

Free option (recommended starting point)

- Azure Custom Vision free tier: up to 10,000 predictions/month and limited monthly training time at no cost. At your expected volume (well under 100 searches/month for a private, non-public site), this comfortably covers ongoing use indefinitely — realistic expected cost: \$0/month.

Paid options, for comparison

- Azure Custom Vision (beyond free tier): roughly \$1.50–2 per 1,000 predictions, plus ~\$3/hour for any training beyond the free allowance — still a few dollars a month at low volume.
- Google Vertex AI (AutoML Vision): no meaningful free tier for hosted inference; bills per node-hour the model is deployed (roughly \$1–1.50/hour) even with no traffic — better suited to steady, higher-volume use than a low-traffic private site.

- AWS Rekognition Custom Labels: similar model to Vertex hourly charge for keeping the endpoint active (several dollars/hour), the least efficient option at low, sporadic volume unless carefully managed.
- Claude API (general vision model, alternative architecture): pay-per-token rather than pay-per-prediction; see Section 7 for a detailed estimate.

Recommendation: start on Azure Custom Vision's free tier. It is purpose-built for training a classifier on your own tagged images, fits a private/low-traffic site precisely, and has no always-on hourly cost like Vertex or Rekognition.

7. Token/Request Cost Estimate — Claude API (Alternative Path)

If a general vision-language model (e.g. Claude) were used instead of a dedicated trained classifier, here's what to expect per request and per month, at an estimated 50 searches/month on a private site:

Approach	Tokens / search	Monthly tokens (×50)	Est. monthly cost
Query photo only (matched against a database, not compared to reference photos)	~1,500 – 2,000	~90,000	< \$1
Query photo + 5 reference photos per search	~8,000 – 10,000	~450,000	~\$1.50 – \$2.50
Query photo + 10 reference photos per search	~15,000 – 16,000	~800,000	~\$3 – \$4

At this volume, even the most token-heavy design stays well under \$5/month — token cost is not the deciding factor between Claude and a dedicated classifier. The deciding factor is accuracy on near-identical locomotives, which a purpose-trained classifier (Section 6) is architecturally better suited to handle, since Claude is a general vision model rather than a model trained specifically on your inventory.

8. What I Need From You

- cPanel/FTP access, or a staging clone of the site (staging preferred for safety while building/testing new pages).
- Database access, or confirmation I can retrieve credentials from Zen Cart's configure.php once inside cPanel.
- Reshoot photos of each locomotive: solid/white background, even lighting, 2–3 angles per item where practical.
- Any known list of visually similar locomotive pairs, so extra training images can be prioritized where they matter most.
- Approval to create the Azure (or alternative) cloud account under your own billing, or confirmation you'd like me to set it up for you to review.
- Confirmation of preferred payment method (Wise / PayPal / bank transfer) and the 50/50 payment split.

9. Limitations & Expectations

- Visual search will not be perfectly accurate for near-identical locomotives (same manufacturer/era/paint) — an inherent limitation of image recognition, not a build defect.
- Low-confidence matches show a short list of likely candidates rather than a single guessed answer.
- Accuracy depends heavily on training photo quality — this is why the reshoot request matters more than any other single factor.
- The existing QR code system remains the primary, reliable lookup method and will not be removed.
- A small proof-of-concept (10–15 locomotives) can be run first to show real accuracy numbers before committing photos for the full collection, if preferred.
- This proposal assumes no major structural issues in the existing Zen Cart codebase; anything unexpected found during discovery will be flagged before further work begins.

10. A Few Final Questions

1. Proof-of-concept on 10–15 locomotives first, or go straight to the full collection?
2. Staging copy of the site, or work directly on the live cPanel environment?
3. Roughly how many total locomotives/items would need training photos?
4. Are you comfortable creating your own Azure account for billing, or would you like it set up for your review first?

Looking forward to your confirmation.

Ahmed Ali

Founder & CEO

Datix AI